

Internship Project Report

Graph Neural Network and Transformer Based Account Takeover Detection

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BONAFIDE CERTIFICATE

This is to certify that the project report entitled **Graph Neural Network and Transformer Based Account Takeover Detection** is a record of the original work carried out by **Komal** under our supervision and guidance. This work is being submitted in partial fulfillment of the requirements for the internship completed at the **Department of Computer Engineering, National Institute of Technology, Kurukshetra**.

To the best of our knowledge, the results embodied in this report have not been submitted to any other University or Institute for the award of any degree ,diploma or certification.

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Abstract

The increasing numbers of Account Takeover (ATO) attacks are becoming a striking threat to digital platforms , where unauthorized peoples gain control over user accounts through compromised login data. Detection of such attacks is becoming very difficult because of two major reasons:the extreme class imbalance between legitimate and takeover events and traditional model failed to capture , relational nature of login events.

This project put forward a hybrid deep learning framework that uses the Graph Attention Network (GATv2) and a Transformer encoder for detecting ATO attacks in good terms. This framework utilized the available Risk-Based Authentication (RBA) login dataset on kaggle. In order to handle structural dependencies , each login event is treated as node and edges are formed on the basis of shared User IDs, IP addresses and Autonomous System Numbers (ASNs). Using this reational modeling technique, we able to detect cooridnated attack patterns.

In order to handle severe class imbalance , an approach is used which utilize majority class undersampling and Synthetic Minority Over-sampling Technique (SMOTE).Training of model is done using cross-entropy loss with label smoothing and evaluated metrics like Precision, Recall, F1-score, Matthews Correlation Coefficient (MCC), and Precision-Recall Area Under Curve (PR-AUC).

The final results shows that integration of structural graph learning and contextual transformer-based representations strongly improves fraud detection performance in extremely imbalanced datasets.This framework emphasize the positive impact of graph-based architectures for real-world cybersecurity applications.

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Chapter 1

Objective

The focus of this project is to draft a deep learning framework for the login authentication systems in order to detect Account Takeover (ATO) attacks. As day by day cyber threats techniques are developing and hence complexity of login dataset as well, therefore this model aims at utilizing relational modeling techniques.

1.1 Primary Objective

The primary objective is to design a hybrid graph-based model that smartly differentiates between legitimate login and account takeover attempts. This framework combines Graph Attention Networks (GATv2) and Transformer Encoder, to capture structural dependencies and contextual feature representations from authentication logs. The goal is to develop a model that can identify abnormal login behavior by usage of shared entities like User IDs, IP Addresses, and Autonomous System Numbers (ASNs).

1.2 Secondary Objectives

In add-on to the primary goal of this model, the project also aims to achieve secondary objectives that are , to use login session data that reflects real-world relational dependencies, in order to construct an entity-relation graph , to analyze the effectiveness of integrating structural learning (GNN) with contextual representation learning (Transformer) and to examine model behavior respect to severe class imbalance conditions.

1.3 Performance Optimization Goals

Since Risk based authentication datasets are typically highly imbalanced , hence the ordinary accuracy-based evaluation is inadequate. Therefore, this project focuses on other metrics which turn out to be more informative in context to severe imbalance dataset. The

performance optimization goals are: Maximizing Precision-Recall trade-off, Achieving high F1-score through threshold evaluation on validation data , Improving Matthews Correlation Coefficient (MCC) to ensure balanced classification performance and minimizing false positives while maintaining strong detection of takeover attempts.

1.4 Research Contribution Goals

In addition to implementation, the project looks for the application of hybrid graph-transformer architectures in cybersecurity applications. The research aims to: Illustrate the effectiveness of relational graph modeling for authentication fraud detection , Provide empirical analysis of GATv2 performance in highly imbalanced datasets, Inspect the role of contextual attention mechanisms in enhancing structural representations and Set up a reproducible pipeline for future research in graph-based cybersecurity systems.

Chapter 2

Introduction

2.1 Background and Evolution of Cyber Threat Landscape

The fast expansion of the digital platforms, cloud-based services, and online financial ecosystems has drastically increased the dependence on authentication systems. Most of the modern applications rely heavily on login-based verification mechanisms to protect the sensitive information like financial, personal and organizational data. However, this growing dependence has simultaneously expanded the attack area available to different adversaries. Cybercriminals increased the exploitations of authentication systems by the help of automated attack techniques, large scale credential leaks and infrastructure to bypass the conventional security controls. Among the cyber threats, *Account Takeover (ATO)* attacks have become one of the most financially damaging attacks. It also cause operational disruption. In an ATO attack, the attacker gains unauthorized access to a user account by using stolen data. These data is obtained using phishing , credential stuffing, malware infections or data breaches. Once the attacker gains the access, they may conduct fake transactions, exfiltrate the sensitive data, or leverage the compromised accounts.

2.2 Challenges in Detecting Account Takeover Attacks

Technical challenges presents in detection process of ATO attacks:

- **Extreme Class Imbalance:** Most of the authentication datasets are heavily bi-ased toward legitimate login events, and takeover events often contributing less than

- **Relational Dependencies:** Login sessions are not independent observations. The coordinated attacks mostly reuse the infrastructure such as IP addresses or ASNs (Autonomous System Numbers), that creates hidden structural relationships between the events.
- **Adaptive Attack Strategies:** Modern attackers copy the legitimate behaviour pattern, making mostly rule-based detection systems increasingly ineffective against modern day attack.

Figure 2.1 Describing lifecycle of an Account Takeover attack, showing how credential gets compromised, which leading to unauthorized access.

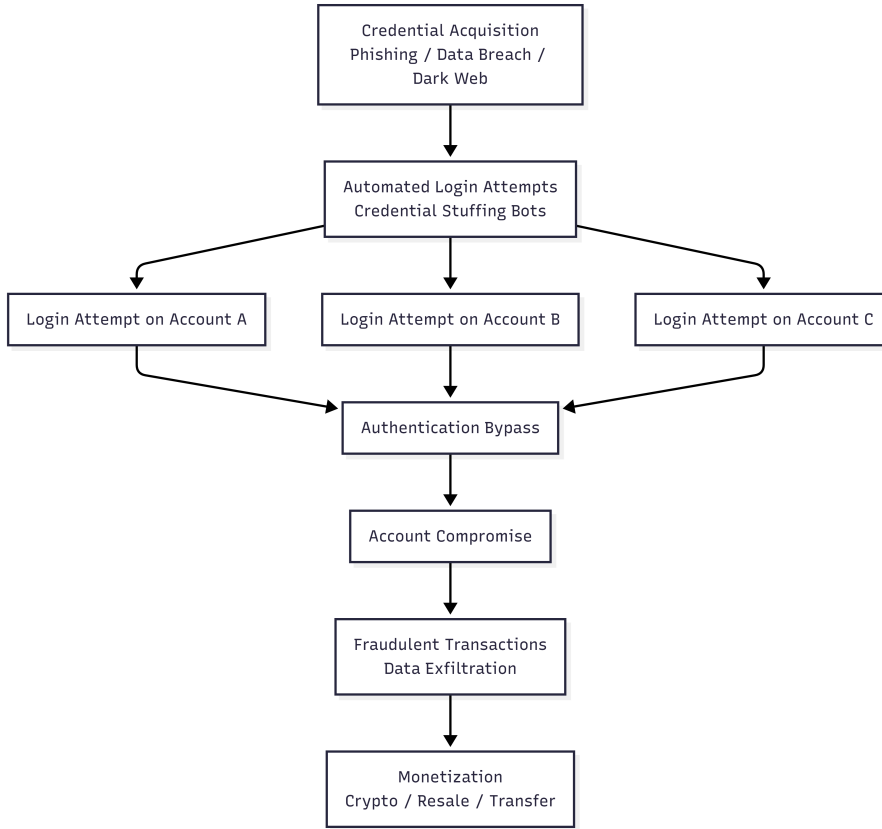


Figure 2.1: Lifecycle of an Account Takeover (ATO) Attack

2.3 Motivation for Graph-Based and Hybrid Learning

Each login attempt treated as an independent event (feature vector) by conventional machine learning models. Such approaches may capture the session-level attributes, they fail to capture the relational dependencies that part of authentication data. Modeling such dependencies requires a graph-based approach instead of conventional methods. Recent

developments in Graph Neural Network (GNNs) have shown strong capability to model relational and structured data [9, 1]. Graph Attention Networks (GAT) enable the nodes to dynamically weight the importance of their neighbours, making them best suited for fraud detection task that involves shared entities. Concurrently, the transformer architectures are revolutionizing the context-based representation using the self-attention mechanisms [8]. Transformers allow models to capture long-term dependencies and complex feature interactions that is beyond local graph neighbourhoods. Above mentioned developments motivated the integration of both structural graph learning with contextual transformer-based representation for the authentication fraud detection.

The main contributions in this projects are as follow:

- Construction of an entity-based authentication graph using shared identifiers.
- Development of a hybrid GATv2–Transformer architecture for relational fraud detection.
- Implementation of imbalance-aware learning using under sampling and SMOTE.
- Validation-based threshold optimization for balanced detection performance.
- Comprehensive evaluation using Precision, Recall, F1-score, MCC, and PR-AUC.

The remainder of this report is structured as follows. The chapter 2 provides a detailed literature review of the existing approaches for fraud detection, graph neural networks and transformer-based models, followed by the detection of research gaps. Chapter 3 describes the methodology, that includes dataset preprocessing, graph construction, and the proposed model architecture. Chapter 4 displays the experimental setup and the evaluation metrics. Chapter 5 discusses the experimental results and analytical discussion. Chapter 6 converse the limitations of the proposed framework. Finally the chapter 7 concludes the report and also highlights the future goals of this.

Chapter 3

Literature Review

The problem of fraud detection and authentication security has grown significantly over the past few years. Early approaches relied only on statistical classification techniques, whereas the modern research combines imbalanced data aware learning, deep neural networks, and graph-based modelling.

In this chapter we will review the existing research across four principal dimensions :

- (i) Imbalance learning
- (ii) evaluation metrics for skewed(biased) data
- (iii) graph neural networks for fraud detection
- (iv) transformer-based contextual modelling

the objective here is to identify or highlight the limitations in current approaches and also highlight the research gaps that are addressed by the proposed framework.

3.1 Imbalance Learning in Fraud Detection

Class imbalance is a never-ending challenge in field of fraud detection and authentication security. In real-world datasets, the fraudulent events or account takeover events represents a very small fraction of the total observations, that ultimately leads to biased datasets resulting in biased learning pattern in conventional classifiers.

Chawla et al. (2002) [2] The Synthetic Minority Over-sampling Technique (SMOTE) , drawing artificial between neighboring minor cases in the feature space in order to build artificial minor classes . In comparison to oversampling by simple duplication, SMOTE increases the size of the minority class decision region which enhances the sensitivity of the classifier. The technique was confirmed in several benchmark data sets and had higher recall than naive oversampling. Nonetheless, SMOTE is based on an assumption that samples are represented in a pure feature space that is of a purely vectorized form and that there are no relational or structural time-dependencies between instances.

Login sessions in authentication systems tend to be interrelated via similar infrastructure like IP addresses or ASNs. Thus, even though SMOTE enhances the balance of

the classes it does not maintain the structural patterns of frauds. To ensure relational integrity SMOTE is utilized along with graph construction in our work.

He and Garcia (2009) [6] A systematic review on learning with imbalanced datasets was purposed in this and the solutions were classified as data level solutions (sampling), algorithm level solutions (cost-sensitive learning), and as hybrid ensemble approaches. Their examination showed that imbalance distorts the border of the decision of the classifiers and deteriorates the ability of detecting minorities.

The authors noted that assessment metrics should be concerned with performance of the minor classes and not the general accuracy. This theoretical basis is what greatly inspires the skewed design we have taken in our authentication detection system.

3.2 Evaluation Metrics in Skewed Classification

In extreme imbalance, performance evaluation is crucial in the fraud detection. Conventional accuracy measures can give false results in the case when majority class takes over the dataset.

Saito and Rehmsmeier (2015) [7] The in-depth analytical comparison between the Receiver Operating Characteristic (ROC) curves and the Precision-Recall (PR) curves was made in their research (saito 2015 precision). Their findings mathematically showed that ROC curves may give excessively optimistic performance estimates when in highly imbalanced environments, and PR curves in general give a more realistic measure of the ability of minority classes to be detected.

This result has a direct impact on the selection of PR-AUC as the main evaluation tool of Account Takeover detection.

Chicco and Jurman (2020) [3] The authors of the paper stated that the Matthews Correlation Coefficient (MCC) offers more balanced measurement of evaluation, as compared to F1-score or accuracy, especially in skewed class distributions (chicco2020mcc). The four elements of the confusion matrix (TP, TN, FP, FN) are taken into consideration in MCC, and it creates a correlation-based performance measure.

As the detection of authentication fraud involves equal evaluation of the positive, as well as the negative prediction, MCC is included in our evaluation framework to supplement Precision-Recall analysis.

3.3 Graph Neural Networks for Fraud Detection

Fraud detection is a relational problem, and the usage of Graph Neural Networks (GNNs) as an interrelational modeling approach has been encouraged.

Veličković et al. (2018) [9]. Graph Attention Networks (GAT) were introduced in this study and They use masked self-attention to estimate the importance of neighbors

by assigning learnable importance weights. GAT enhances the learning of representations using graph that dynamically weight neighboring contributions.

Even though attention mechanism was initially tested on datasets of citation networks, it is especially applicable to fraud detection, where some connections (e.g., shared IP addresses) can be stronger predictors than others. Nevertheless, attention coefficients in the original formulation of GAT are calculated in a steady state.

Brody et al. (2021) [1] overcame this limitation by proposing GATv2 that redefines attention computation, allowing dynamic estimation of importance. They found that their study had a better expressiveness and theoretical benefits in comparison to a fixed attention mechanism.

This development is the direct reason to use GATv2 in our proposed architecture to capture better dynamic relational dependencies in authentication graphs.

Weber et al. (2019) [10] Graph Convolutional Networks (GCNs) were implemented by Weber et al. (2019) to detect anti-money laundering on Bitcoin networks. Experimental evidence provided by them demonstrated that graph-based models are better than traditional feature-based classifiers in terms of structural patterns of transactions. Nevertheless, they only used graph convulsion without the contextual transformer-based feature interaction modeling.

It means that hybrid architectures provide a possibility to improve the performance of fraud detection even more.

Hamilton (2020) [5] Theoretical backgrounds of graph representation learning were also presented by this paper , it includes message-passing mechanisms and neighborhood aggregation strategies. The graph construction strategy adopted in our authentication system, which makes use of the framework outlined in this work, is grounded in the notion that the logic of generating the login sessions should be laid out as nodes connecting through the shared identifiers.

3.4 Deep Learning Approaches in Fraud Detection

Other than graph modeling, deep neural networks have been used in the detection of fraud.

Fiore et al. (2019) [4] (cited in the article titled Using Generative Adversarial Learning, 2019) The authors of investigated the effectiveness of applying the concept of Generative Adversarial Networks (GANs) in order to enhance the performance of fraud classification algorithms by creating synthetic minority samples.

Their findings showed that generative models are able to increase the representation of minority classes and increase F1-score. Nonetheless, the method, like conventional imbalance methods, works on tabular feature space, without explicitly modeling the relationships between instances.

3.5 Transformer-Based Contextual Modeling

Vaswani et al. (2017) [8] The attack on the recurrence of long-range dependencies introduced the Transformer architecture which uses self-attention as a method of capturing these dependencies. The attention mechanism calculates combination of the input representations weighted, hence feature interactions of context can be used to model interactions.

Though the paradigm of self-attention is originally aimed at processing natural language, it is quite relevant to authentication information where the interaction of features can indicate abnormal behavior. Nonetheless, the information about graph topology is not implemented in transformers, which is why it is advisable to combine them with graph neural networks in our hybrid framework.

3.6 Comparative Analysis of Existing Work

Table 3.1: Detailed Comparative Analysis of Related Work

Author	Dataset Used	Primary Metrics	Key Results	Identified Limitations
Chawla et al. (2002)	UCI benchmark datasets	Accuracy, Recall	Improved minority recall using synthetic oversampling	No relational or structural modeling
He & Garcia (2009)	Survey of imbalance datasets	Accuracy, ROC	Comprehensive analysis of imbalance challenges	No experimental fraud dataset
Saito & Rehmsmeier (2015)	Binary classification benchmarks	PR-AUC, ROC-AUC	Demonstrated PR superiority in skewed datasets	Not domain-specific to fraud
Chicco & Jurman (2020)	Bioinformatics datasets	MCC, F1-score	MCC provides balanced evaluation under skew	Evaluation-focused study
Veličković et al. (2018)	Citation networks (Cora, PubMed)	Node accuracy	Introduced attention-based graph learning	Static attention mechanism
Brody et al. (2021)	Graph benchmark datasets	Accuracy	Dynamic attention improves expressiveness	No fraud application
Weber et al. (2019)	Bitcoin transaction graph	ROC-AUC	Graph convolution improved AML detection	No contextual transformer modeling
Hamilton (2020)	Theoretical framework	–	Formalized graph representation learning	Conceptual; no dataset evaluation
Fiore et al. (2019)	Credit card fraud dataset	F1-score, Recall	GAN improved fraud detection performance	No relational graph modeling
Vaswani et al. (2017)	NLP translation datasets	BLEU score	Introduced self-attention transformer	No graph topology modeling

3.7 Identified Research Gaps

Based on the reviewed literature, the following research gaps are identified:

- **Feature-Space Imbalance Handling:** Majority of imbalance learning methods like SMOTE and cost-sensitive learning are purely in tabular feature space. Such approaches enhance the representation of minorities but fail to consider structural relationships between authentication events.
- **Limited Integration of Structural and Contextual Learning:** Graph-based fraud detection models are effective in terms of capturing relational dependencies, but generally only utilise neighbourhood aggregation methods. They take little to no cognizance of advanced contextual modeling schemes like transformer-based self-attention.
- **Lack of Graph Topology in Transformer Models:** Transformer architectures give strong contextual representation learning but they do not fundamentally encode the graph topology or shared-entity relationships that are present in the authentication data.
- **Underexplored Authentication-Based ATO Detection:** Much of the previous studies are dedicated to fraud of financial transactions (e.g., credit card or cryptocurrency networks). The use of entity-relation graph to detect login-based Account Takeover has been relatively under explored.
- **Inadequate Emphasis on Imbalance-Specific Metrics:** Despite a number of studies emphasizing the need of Precision-Recall analysis and MCC, most of the deployed systems of fraud detection use accuracy or ROC-AUC that can give misinterpretations when there is extreme imbalance between classes.

It is these gaps that drive the creation of an integrated framework comprising of imbalance-aware sampling, entity-relation graph-building, dynamic graph attention (GATv2), and transformer-based contextual encoding that can be used to perform ATO detection through authentication.

Chapter 4

Proposed Methodology

4.1 Model Overview

The framework is designed to detect Account Takeover (ATO) attacks by combining imbalance-aware sampling, graph-based structural modeling, and contextual representation learning. The overall pipeline of the system is shown in Figure 4.1

This model uses Risk-Based Authentication (RBA) dataset available on Kaggle. Although I didn't utilize complete dataset, only most relevant attributes are selected to reduce computational overhead and to get more generalized results.

And then the selected attributes undergo preprocessing which includes missing value handling, logarithmic transformation of latency, and encoding of high-cardinality categorical identifiers. On the other hand to handle extreme class imbalance present in chosen dataset, we applied majority class undersampling and Synthetic Minority Over-sampling Technique (SMOTE) for generating synthetic minority samples.

After that, balanced dataset is used to create an entity-relation graph in which each login event is treated as a node and events that share common User ID, IP address, or Autonomous System Number (ASN) are treated as edges. Also on the basis of strength of entity relationship, edge weights are assigned.

Now the graph-structured data is feed to a hybrid deep learning architecture which consists GATv2 layers and transformer encoder. A residual fusion mechanism also implemented before final classification. The training of model is done using cross-entropy loss with label smoothing and adam optimizer. F1 score is maximized by using validation-based threshold selection strategy.

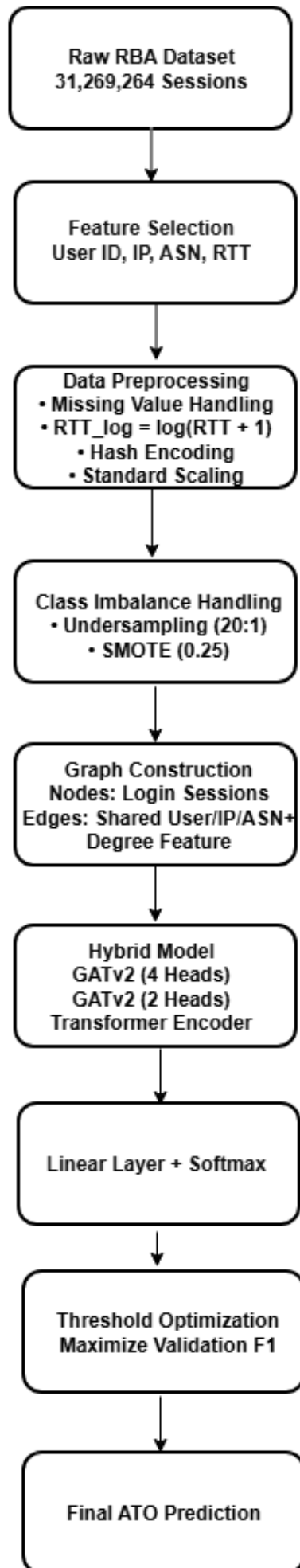


Figure 4.1: architecture of the ATO detection framework.

As shown in Figure 4.1, the framework follows a pipeline that begins with dataset preparation and imbalance correction, then entity-relation graph construction and hybrid deep learning classification. Each step in this pipeline is designed to continuously improve structural and contextual representation.

4.2 Dataset Description

The model uses the *Login Data Set for Risk-Based Authentication (RBA)* available on Kaggle. This dataset contains real-world login attempts that are labeled as legitimate sessions or account takeover events, this dataset is developed by the incident response team of an online service provider.

The original dataset contains multiple attributes that are related to network information, device characteristics, geolocation, and authentication metadata. A overview of the original dataset attributes is given below in Table 4.1.

Table 4.1: Original RBA Dataset Attributes

Feature	Data Type	Description
IP Address	String	Client IP address
Country	String	Country derived from IP
Region	String	Region derived from IP
City	String	City derived from IP
ASN	Integer	Autonomous System Number
User Agent String	String	Browser and device string
OS Name and Version	String	Operating system information
Browser Name and Version	String	Browser details
Device Type	String	Device category
User ID	Integer	Unique account identifier
Login Timestamp	Integer	Timestamp of login attempt
Round-Trip Time (RTT) [ms]	Integer	Network latency in milliseconds
Login Successful	Boolean	Login success indicator
Is Attack IP	Boolean	Known malicious IP indicator
Is Account Takeover	Boolean	Ground-truth takeover label

4.2.1 Selected Features for Modeling

As the original dataset contains multiple attributes, but this model focuses on features that positively aligned with our objectives. Following are the attributes that were selected:

- User ID
- IP Address
- Autonomous System Number (ASN)

- Round-Trip Time (RTT) [ms]

The target variable is:

- Is Account Takeover (Binary: False = Legitimate, True = Takeover)

These features were selected because they let us model the relational dependencies between login events through shared identifiers. other attributes are excluded in order to reduce dimensionality and computational overhead , also to focus on entity-based graph construction.

4.2.2 Class Distribution and Imbalance

Generally , the real world authentication dataset shows a severe class imbalance pattern, in which the numbers of legitimate logins become majority and the account takeover attempts become minority. Dataset revealed a highly skewed distribution at the phase of initial analysis.

This imbalance becomes a challenge for standard classification models, hence accuracy may become misleading because model fails to detect minority class events. To tackle this problem , the implementation of imbalance-aware sampling techniques id done , which is described in Section 3.4.

4.2.3 Statistical Overview

The original RBA dataset contains of 31,269,264 login sessions. Out of these , 31,269,123 sessions are legitimate logins, whereas only 141 sessions are as Account Takeover (ATO) events.

This results in an extreme class imbalance ratio of approximately:

$$1 : 221,766$$

where for every 1 takeover event, more than 221,000 legitimate sessions exist.

Such kind of scenarios make model impractical and computationally not feasible.and hence accuracy based evaluation become misleading .

4.3 Data Preprocessing

In order to achieve stable model training and reliable results , multi step preprocessing is done , before graph contruction.

4.3.1 Missing Value Handling

During the process of feature transformation, missing values handled using different ways to prevent null propagation.

- Round-Trip Time (RTT) filled using median where missing.
- ASN values replaced with a placeholder identifier.
- User ID values assigned a placeholder identifier where missing.
- IP addresses were filled with a default value.

Median imputation was selected for RTT to avoid distortion caused by extreme latency outliers.

For RTT , Median is choosen to avoid issues caused by extreme latec

4.3.2 Logarithmic Transformation

since Network latency values nearly exhibit strong right-skewness. To stabilize variance and compress extreme values, logarithmic transformation was applied:

Logarithmic transformation is applied, because here network latency values shows strong right-skeness therefore to stabilize variance and to compress extreme values .

$$RTT_{log} = \log(RTT + 1) \quad (4.1)$$

Here this addition of 1 maintains numerical stability to handle zero-valued entries.

4.3.3 Categorical Encoding

Hash-based transformations used to handle high-cardinality categorical attributes such as User ID and IP Address. With this approach, model can

- Prevents dimensional explosion from one-hot encoding
- Preserves relative uniqueness
- Ensures memory-efficient representation

For ASN to encode , used categorical index conversion.

4.3.4 Feature Scaling

z-score normalization used to standardized our numerical features before graph modelling,

$$X_{scaled} = \frac{X - \mu}{\sigma} \quad (4.2)$$

where μ represents the mean and σ the standard deviation.

Standardization helps to balance gradient updates and also improves convergence stability in deep graph networks.

4.4 Class Imbalance Handling

A two-phase balancing technique is used to handle extreme imbalance ratio (1:221,766), because this imbalance making the model impractical.

4.4.1 Undersampling Strategy

Before synthetic oversampling , the majority class i.e. legitimate events are randomly undersampled.

The number of legitimate samples :

$$N_{majority} = 20 \times N_{minority} \quad (4.3)$$

Now , imbalance ratio becomes 20:8 approx

Undersampling provides two major purposes:

- Prevents graph size explosion
- Controls memory consumption
- For training , it maintains sufficient negative samples

4.4.2 Synthetic Minority Over-Sampling Technique (SMOTE)

After done with undersampling, SMOTE [2] was used to generate synthetic minority samples.

by doing interpolation between a minority sample and its nearest neighbor , SMOTE generates synthetic instances

$$x_{new} = x_i + \lambda(x_{nn} - x_i) \quad (4.4)$$

where:

- x_i is a minority sample
- x_{nn} is one of its nearest neighbors
- $\lambda \in (0, 1)$ is a random scalar

Here , the sampling strategy was set to 0.25 , that indicates that the minority samples increased to 25% of majority class size.

4.4.3 Sampling Ratio Justification

Since , real world authentication systems very rarely gets equal class distribution therefore a perfectly balanced 1:1 dataset is avoided. Maintaining partial skew:

- Preserves real-world fraud characteristics
- Prevents overfitting to synthetic patterns
- Maintains realism in operations

4.5 Graph Construction

After handling of imbalance dataset , login sessions converted into graph representation to model relational dependencies.

4.5.1 Node Definition

Each login session is a node in the graph.

The node feature vector is :

$$X_i = [User_{enc}, ASN_{enc}, IP_{enc}, RTT_{log}, Degree] \quad (4.5)$$

where Degree represents structural connectivity of node.

4.5.2 Edge Construction Strategy

Edges were created between those nodes that share common identifiers:

- Shared User ID (weight = 1.0)
- Shared IP Address (weight = 0.7)
- Shared ASN (weight = 0.4)

To prevent quadratic growth in dense groups, a maximum of 15 connections per entity group was enforced.

Here, a maximum of 15 connections per entity group is set in order to prevent quadratic growth in dense groups.

4.5.3 Degree as Structural Feature

Node degree was computed as:

$$Degree_i = \frac{deg(i)}{\max(deg) + \epsilon} \quad (4.6)$$

where $deg(i)$ represents the number of connections of node i .

Coordinated infrastructure reuse or attack clusters, may be indicated by high-degree nodes

4.5.4 Graph Formulation

The authentication graph is defined as:

$$G = (V, E) \quad (4.7)$$

where:

- V represents login sessions
- E represents entity-based weighted connections

4.6 Model Architecture

The model integrates Graph Attention Networks (GATv2) with a Transformer encoder to jointly learn structural and contextual representations from authentication graphs. the architecture contains two GATv2 layers followed by a Transformer encoder. Then a residual fusion mechanism also used to combine structural and contextual representations before final classification.

This model integrates Graph Attention Networks (GATv2) and Transformer encoder to learn structural and contextual representations.

4.6.1 Graph Attention Network (GATv2)

In order to capture relational dependencies between login events, two GATv2 layers [9, 1] used.

The first GATv2 layer uses 4 attention heads with 64 output features per head. The second GATv2 layer uses 2 attention heads with 64 output features per head.

Dynamic importance weights between connected nodes is computed by graph attention . The attention coefficient between node i and node j is defined as:

$$\alpha_{ij} = \text{softmax}_j \left(a^T [Wh_i \parallel Wh_j] \right) \quad (4.8)$$

where:

- h_i and h_j are node feature vectors,
- W is a learnable weight matrix,
- a is a learnable attention vector,
- $\parallel$ denotes concatenation.

To enhance representation stability , multi-head attention mechanism is used and also to prevent overfitting , Dropout with the rate 0.3 used .

4.6.2 Transformer Encoder

Transformer encoder [8] was used after GAT layers , to model higher-order feature interactions over local graph neighborhoods.

The encoder configuration includes:

- Model dimension: 128
- Number of attention heads: 4
- Feedforward dimension: 256
- Number of layers: 1

To capture global contextual relationships , Self-attention mechanism used. Transformer encoder uses feature wise attention to refine representations while normal graph attention focuses on neighborhood level aggregation.

4.6.3 Residual Fusion Mechanism

Here residual fusion applied to preserve structural information that learned by GAT layers,

$$X_{final} = X_{gnn} + X_{transformer} \quad (4.9)$$

Residual connections help model to improve gradient flow, stabilize deep training, and prevent representation degradation.

4.6.4 Classification Layer

These fused representation is passed through a fully connected linear layer whose mapping is 128 features to 2 output logits:

then

$$Z = WX_{final} + b \quad (4.10)$$

Conversion of logits into class probabilities representing legitimate and takeover predictions by Softmax activation.

4.7 Training Strategy

4.7.1 Loss Function

For binary classification, the cross-entropy loss is defined as:

$$L = -(y \log(p) + (1 - y) \log(1 - p)) \quad (4.11)$$

To reduce overconfidence and to improve generalization , Label smoothing with factor 0.05 is used .

4.7.2 Optimizer

The Adam optimizer was used with learning rate of 0.0008. Its adaptive learning capability and stable convergence behavior for graph neural networks makes Adam selection suitable here.

4.7.3 Early Stopping

Then Early stopping was implemented with patience of 20 epochs. When validation F1-score failed to improve , training will be stopped.

Here , F1 score selected to monitore metric instead of accurcay due to the imbalanced nature of dataset.

4.7.4 Threshold Optimization

Threshold optimization was performed using validation data.

The threshold was searched in the range:

$$t \in [0.1, 0.9] \quad (4.12)$$

The value maximizing validation F1-score was selected as the optimal decision threshold.

IN this way , it ensures improvement in minority detection performance under imbalanced conditions.

Chapter 5

Experimental Results and Analysis

5.1 Experimental Setup

Here in this section , the description of computational environment, software configuration and implementation details is mentioned .

5.1.1 Hardware Configuration

The model is coded in Google Colab environment with following specifications: GPU: NVIDIA Tesla T4 (if available) ,RAM: 12–16 GB and Runtime: Python 3.x .The usage of GPU acceleration helped to reduce training time during optimization process of graph neural network.

5.1.2 Software Environment

The implementation was done using the following libraries:PyTorch,PyTorch Geometric,Scikit-learn,Imbalanced-learn (SMOTE), NumPy, Polars (for efficient CSV handling),Matplotlib and Seaborn (for visualization). To ensure reproducibility, model is implemented in Python using modular training pipeline

5.2 Evaluation Metrics

As the dataset is highly imbalanced in nature , therefore multiple important evaluation metrics used here , mentioned below:

5.2.1 Precision

Precision measures the proportion of predicted takeover events that are actually correct.High precision indicates low false positive rate.

$$Precision = \frac{TP}{TP + FP} \quad (5.1)$$

Here,

- TP = True Positives
- FP = False Positives

5.2.2 Recall

Recall measures the proportion of actual takeover events correctly detected. High recall ensures that fewer malicious events missed.

$$Recall = \frac{TP}{TP + FN} \quad (5.2)$$

Here,

- TP = True Positives
- FN = False Negatives

5.2.3 F1-Score

F1-score represents the harmonic mean of precision and recall. F1 score provides a balanced measure in imbalanced classification problems.

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (5.3)$$

5.2.4 Matthews Correlation Coefficient (MCC)

Even when class distributions are skewed, MCC provides a balanced evaluation. MCC ranges from -1 to 1, where 1 indicates perfect prediction.

$$MCC = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad (5.4)$$

5.2.5 Precision–Recall Area Under Curve (PR-AUC)

The area under the precision–recall curve across varying thresholds is measured by PR-AUC. As it focuses on minority class detection performance, hence it is more informative than ROC-AUC for highly imbalanced datasets.

5.3 Results on Training Data

Table 5.1: Training Performance Metrics

Metric	Value
Precision	0.9036
Recall	0.9378
F1-score	0.9204
MCC	0.9002
PR-AUC	0.8869

These training results shows strong detection capability, with an F1-score of 0.9204 and MCC of 0.9002, this indicates balanced performance across both classes. The high recall (0.9378) shows that model successfully captures the majority of takeover events, while precision (0.9036) shows controlled false positive predictions. The PR-AUC of 0.8869 confirms strong minority class discrimination during training.

5.4 Results on Test Data

Table 5.2: Test Performance Metrics

Metric	Value
Precision	0.8579
Recall	0.9261
F1-score	0.8907
MCC	0.8632
PR-AUC	0.8240

Even on unseen test data, the model kept strong generalization performance. The F1-score of 0.8907 and MCC of 0.8632 indicate that the model maintain balanced predictive power over training data. Recall remains high at 0.9261, this value of recall shows that most takeover attempts are detected. Although precision slightly decreases to 0.8579, but still it remains within acceptable operational limits. The PR-AUC of 0.8240, this value confirms effective minority class detection under imbalanced conditions.

5.5 Confusion Matrix Analysis

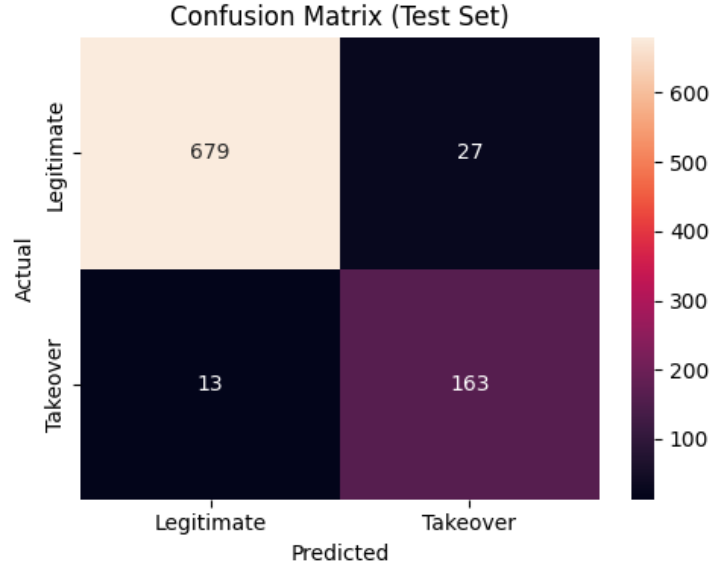


Figure 5.1: Confusion Matrix on Test Set

The confusion matrix in Figure 5.1 provides detailed overview of classification behavior. This model correctly identifies 163 takeover events (True Positives) , on the other hand it only misclassified 13 malicious sessions as legitimate (False Negatives). The low false negative count is more important , as missed attacks may lead to big issues. Out of legitimate sessions, 679 correctly classified (True Negatives), while 27 incorrectly flagged as takeover (False Positives). False positives may cause minor inconvenience, but they are very less severe in comparison to false negatives . But overall , the confusion matrix indicates towards a well-balanced detection system.

5.6 Precision–Recall Curve Analysis

The Precision–Recall curve shown in Figure 5.2 shows the trade-off between precision and recall over varying decision thresholds. This PR-AUC score of 0.8240 clearly indicates a strong minority class discrimination. As PR-AUC mainly focuses on positive class performance , which makes this a more suitable for highly imbalanced datasets in comparison to ROC-AUC. By looking at the curve , we can conclude that stable precision across a wide recall range , which indicates threshold optimization and nicely balances false positives and false negatives.

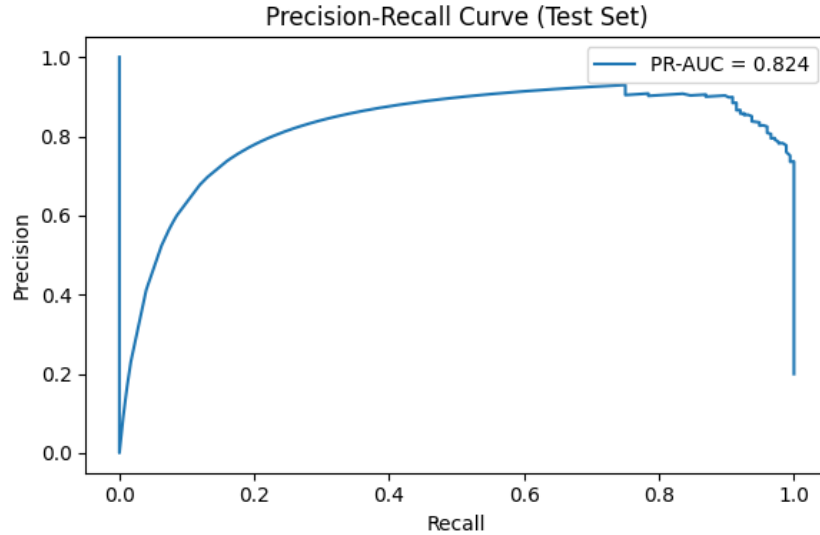


Figure 5.2: Precision–Recall Curve on Test Set

5.7 Training Loss Analysis

The training loss curve in Figure 5.3 shows stable convergence behavior. During early epochs, the loss decreases consistently and stabilizes itself before early stopping is triggered. Early stopping based on validation F1-score helps to prevent overfitting and it also ensures better generalization. Stable gradient updates during optimization is result of absence of sharp oscillations.

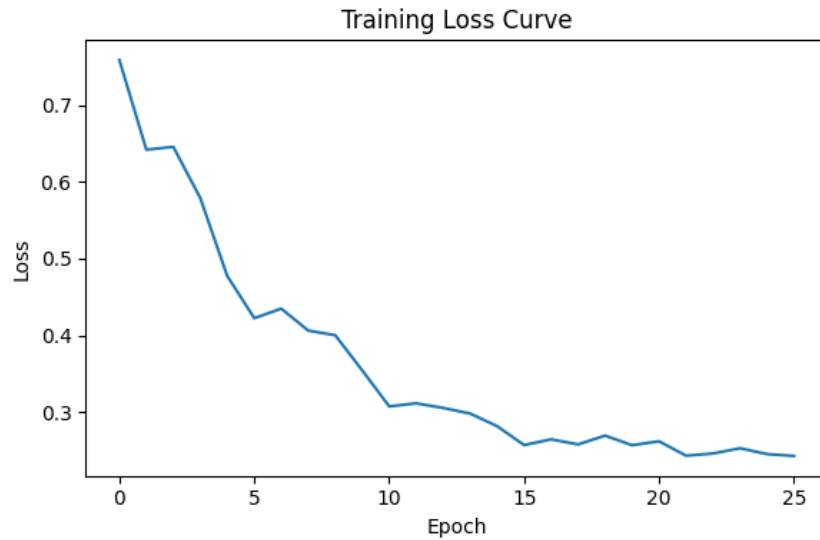


Figure 5.3: Training Loss Curve

Chapter 6

Discussion

6.1 Performance Interpretation

This framework shows strong and balanced detection capability over both training and test datasets. The test F1-score of 0.8907 and MCC of 0.8632 indicate strong classification performance under imbalanced conditions. Here a key observation is the high recall value (0.9261) on test set, which shows that majority numbers of account takeover is successfully detected. In authentication system, minimizing false negatives is more important as missed attacks can result in big issues. Although here precision slightly decreases compared to training performance, but still it remains within acceptable operational limits. The small gap between training and test F1-scores (approximately 0.03) indicates minimal overfitting and strong generalization.

6.2 Why Accuracy is Misleading in Fraud Detection

Since there is extreme class imbalance in the RBA dataset, that's why accuracy intentionally not used as a primary evaluation metric. Here, imbalance ratio 1:221,766 in the original dataset approx, a naive classifier will achieve near perfect accuracy while failing to detect takeover events in this situation. In highly imbalanced datasets, the large number of true negatives dominates calculation, masking poor minority class performance. For example, if all samples classified as legitimate: TN would be extremely high, TP would be zero and Accuracy would still appear artificially high. Therefore, evaluation metrics such as Precision, Recall, F1-score, MCC, and PR-AUC were prioritized in this model for evaluation, as they provide meaningful and reliable results.

6.3 Strengths of the Proposed Model

The proposed hybrid framework offers several advantages: **Structural Awareness:** Through shared User IDs, IP addresses, and ASNs, Entity-relation graph construction captures coordinated behavior. **Dynamic Attention Modeling:** Adaptive importance weighting of neighboring login sessions is enabled by GATv2 layers. **Contextual Feature Interaction:** The Transformer encoder refines node embeddings through self-attention, which enables higher-order feature interactions. **Imbalance-Aware Optimization:** Here Two-step balancing technique combined with threshold optimization helps to improve minority detection performance. **Balanced Evaluation:** Use of MCC and PR-AUC ensures reliable performance under skewed class distributions. The combination of structural and contextual learning components eventually contributes to improved detection of authentication attacks.

6.4 Identified Limitations

Despite strong performance, several limitations are observed: **Static Graph Construction:** The authentication graph is designed as static structure. Also temporal evolution of login behavior is not explicitly modeled. **Limited Feature Utilization:** Here Only identity and network-level attributes used. Device fingerprints, geolocation drift, and behavioral signals excluded. **Synthetic Oversampling Dependency:** SMOTE-generated samples may not perfectly represent real-world attack patterns scenarios. **Single Transformer Layer:** To further enhance contextual modeling, A deeper transformer architecture can be utilized. **Offline Evaluation:** The model was evaluated in batch setting and hence not tested under real-time streaming conditions. These limitations provide potential paths for future research and system enhancement.

Chapter 7

Conclusion

This work presented a hybrid graph-transformer framework that detect Account Takeover (ATO) attacks in authentication systems. This study also addressed many challenges, particularly extreme class imbalance and structural dependency between login events, that generally exist in real world authentication dataset. The original Risk-Based Authentication (RBA) dataset holds a severe imbalance ratio approx 1:221,766 which makes accuracy based evaluation misleading and training infeasible. To handle this challenge, a two step imbalance handling technique applied, which combines majority undersampling and SMOTE-based minority oversampling. The core of this work is modeling authentication sessions as an entity-relation graph. Here, didn't treated login attempts as independent records, while structural dependencies were captured through shared User IDs, IP addresses, and Autonomous System Numbers (ASNs). This technique enabled model to identify those patterns that often associates with takeover attacks. Then the hybrid architecture integrated two GATv2 layers for structural attention learning with a Transformer encoder for contextual feature refinement. Residual fusion also used to ensure stable representation integration. For further improvement, validation-based threshold optimization applied. These results demonstrate strong and balanced performance. On the test dataset, the model achieved: F1-score: 0.8907, MCC: 0.8632, Recall: 0.9261, PR-AUC: 0.8240. The high recall shows effective detection of takeover events, while the strong MCC also confirms balanced predictive capability. The small gap between training and test performance suggests good generalization without significant overfitting. Overall, the results indicates that combining structural graph modeling with contextual transformer learning enhances authentication fraud detection compared to traditional approaches.

However, the final framework shows strong results, but still some enhancements can be explored in future. First, To capture sequential evolution of login behaviour, temporal modeling can be utilized. the time-aware attention mechanisms or Dynamic graph neural networks may improve detection more efficiently. Second, To enrich contextual representation, additional behavioral and device level features for example browser fingerprinting, session duration patterns, can be integrated. Third, can replace the static

SMOTE oversampling with generative adversarial models , then it can produce more realistic synthetic attack samples.Fourth, deeper transformer architectures or multi-layer attention mechanism help to enhance higher-order feature interaction modeling.Fifth,To evaluate inference latency and operational feasibility,real-time deployment under streaming authentication environments can be explored. Finally, new and modern AI techniques for GNN can be explored to improve interpretability , this also allows security analysts to understand relational factors that can further contribute to takeover detection decisions.These points can provide many opportunities to do further advance research in this domain .

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